1 **(a)** The internal energy of an ideal gas is just the sum of the microscopic kinetic energy, due to the random motion of its molecules since the microscopic potential energy of an ideal gas is zero.

Its internal energy depends only on its state (pressure, volume, temperature) and amount of gas.

(b) **(i)** **1.** $\Delta U = Q + W$

Since volume is constant, $W = 0$.

$$
\begin{aligned}
Q &= \Delta U \\
&= \frac{3}{2}p_2V - \frac{3}{2}p_1V \\
&= \frac{3}{2}V\left(p_2 - p_1\right) \\
&= \frac{3}{2}\left(2.0\times10^{-2}\right)\left(1.5\times10^{5} - 1.0\times10^{5}\right) \\
&= 1500.0 = 1500 \text{ J (shown)}
\end{aligned}
$$

2. Since

$$E_k = \frac{3}{2}kT_1 = 6.2\times10^{-21} \text{ J}$$

$$T_1 = \frac{2}{3}\left(\frac{6.2\times10^{-21}}{1.38\times10^{-23}}\right) = 299.52 \text{ K}$$

From $p = \frac{nR}{V}T$, since volume is constant,

$$\Delta p = \frac{nR}{V}\Delta T = \frac{p_1}{T_1}\Delta T$$

$$
\begin{aligned}
\Delta T &= \frac{\Delta p}{p_1}T_1 \\
&= \frac{0.5\times10^{5}}{1.0\times10^{5}}(299.52) \\
&= 149.76 = 150 \text{ K}
\end{aligned}
$$

OR

$$E_k = \frac{3}{2} k T_1$$

$$T_1 = \frac{2}{3} \frac{E_k}{k}$$

Since volume is constant,

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$T_2 = \frac{p_2}{p_1} T_1 = \frac{p_2}{p_1}\left(\frac{2}{3}\frac{E_k}{k}\right)$$

$$\begin{aligned}
T_2 - T_1 &= \frac{p_2}{p_1}\left(\frac{2}{3}\frac{E_k}{k}\right) - \frac{2}{3}\frac{E_k}{k} \\
&= \frac{2}{3}\frac{E_k}{k}\left(\frac{p_2}{p_1} - 1\right) \\
&= \frac{2}{3}\left(\frac{6.2\times10^{-21}}{1.38\times10^{-23}}\right)\left(\frac{1.5\times10^{5}}{1.0\times10^{5}} - 1\right) \\
&= 149.76 = 150 \text{ K} = 150 \text{ °C}
\end{aligned}$$

OR

$U_1 = NE_k$ where N is the number of gas molecules

$$N = \frac{U_1}{E_k}$$

$$\begin{aligned}
\Delta U &= U_2 - U_1 \\
&= \frac{3}{2} NkT_2 - \frac{3}{2} NkT_1 \\
&= \frac{3}{2} Nk\left(T_2 - T_1\right)
\end{aligned}$$

$$\begin{aligned}
T_2 - T_1 &= \frac{2}{3}\frac{\Delta U}{Nk} \\
&= \frac{2}{3}\frac{\Delta U}{\left(U_1/E_k\right)k} \\
&= \frac{2}{3}\frac{E_k \Delta U}{\left(\frac{3}{2} p_1 V\right)k} \\
&= \frac{4}{9}\frac{\left(6.2\times10^{-21}\right)\left(1500\right)}{\left(1.0\times10^{5}\right)\left(2.0\times10^{-2}\right)\left(1.38\times10^{-23}\right)} \\
&= 149.76 = 150 \text{ K} = 150 \text{ °C}
\end{aligned}$$

(ii) The first law of thermodynamics states that the increase in internal energy ΔU is equal to the sum of the work done on the system W and the heat supplied to the system Q, i.e. $\Delta U = Q + W$. Hence $Q = \Delta U - W$.

When a unit mass of the gas, heated under constant volume or under constant pressure, experiences a unit rise in temperature, ΔU will be the same since $\Delta U \propto \Delta T$.

When the gas is heated at constant volume, there is no work done. W is zero and $Q_V = \Delta U$.

When the gas is heated at constant pressure, work is done by the gas as it expands. W is negative and $Q_p > \Delta U$.

The specific heat capacity of a gas is the heat supplied to a unit mass of the gas to cause a unit rise in its temperature i.e. $c = \dfrac{Q}{m(\Delta T)}$. Since $Q_p > Q_V$ the specific heat capacity at constant pressure is higher than that at constant volume.

Raffles Institution
Year 5-6 Physics Department

2 **(a)** Since *k* and *m* are constant, $a \propto x$